

October 2026 action-derived source, exact ACROPOLIS and independent-cascade update

Planck PR4/NPIPE, GALAH, external covariance and cascade qualification retained

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EVIDENCE STATUS: Formal paper update. The Planck, GALAH, 10,000-draw fluence, exact ACROPOLIS, action-history bridge and targeted independent-cascade calculations are technically recorded.

SCIENTIFIC STATUS: Gate incomplete. The exact cascade admits conditional abundance-compatible parameter space, but the minimal action underproduces the mediator population, the full independent-cascade basin does not close, and ordinary D-chain uncertainty remains strongly disfavoured.

PROMOTION BOUNDARY: no lithium solution or discovery is claimed. Promotion requires a frozen production action, exact external D-chain and mass-7 posterior products, an external coupled cascade or independent nuclear response, and full FIRAS/ACT/SPT consistency.

Description

The early Universe made mostly hydrogen and helium, plus tiny amounts of deuterium, helium-3 and lithium.

Modern calculations predict deuterium and helium very well, but they predict too much lithium compared with old metal-poor stars.

This long-standing mismatch is termed the cosmological lithium problem.

The July tests tried broad ways to change the early-Universe calculation.

Those did not work: when lithium moved in the right direction, something else, usually deuterium or helium, moved out of agreement.

The August work then tested a more selective idea: a late, narrow electromagnetic process could reduce the mass-7 reservoir after normal nucleosynthesis without disturbing the successful deuterium and helium predictions.

The September update asks whether this selected candidate is still acceptable when compared with the detailed Planck CMB data, including calibration and foreground nuisance parameters.

The current answer is yes for the CMB gate, but the later nuclear-rate checks sharpen the boundary: the candidate remains conditional on D/H, lithium, helium and helium-3 behaviour under realistic nuclear-rate covariance and direct mediator-aware abundance propagation.

Latest Results - Overview through 2 October 2026

The GALAH DR3/DR4 analysis retains a censored stellar-surface target of $A(\text{Li}) = 2.09062$, corresponding to $\text{Li}/\text{H} = 1.232 \times 10^{-10}$, with a 95% interval of $1.073\text{--}1.424 \times 10^{-10}$. It remains an empirical stellar anchor rather than a direct primordial-abundance measurement.

The later programme moved beyond the September rate-pull search. A 10,000-draw fluence-matched PRIMAT run, convergence-controlled LINX propagation and an exact 15,768-row ACROPOLIS decay scan established conditional abundance-compatible parameter space. Fifty-eight ACROPOLIS rows passed every registered D/H branch; the best had $\text{Li7}/\text{H} = 1.2383 \times 10^{-10}$ and abundance $\chi^2 = 0.9045$.

The physical interpretation is narrower. The minimal scalar action did not generate enough late source, while a calibrated parent-decay bridge and the best independent reduced cascade passed technically. The full 58-row independent basin did not close, although explicit pair production and active-band photon redistribution reduced the selected discrepancies to below 1.588%. Ordinary nuclear-rate error remains disfavoured, and the remaining mass-7 boundary is unresolved pending external posterior products.

Connection to Existing Papers

This revision follows the stronger Physics Reports style in which external papers are integrated into the structure of the argument. The references below are not simply a reading list.

Each one defines either an observational anchor, a numerical-method benchmark, a physical safety gate or a boundary on what can be claimed.

The most important model for this structure is the 2026 BBN sensitivity-atlas review by Burns, together with its public atlas repository.

That work separates BBN into observational inputs, theoretical machinery, sensitivity coefficients, uncertainty budgets and illustrative applications.

The present BBN-Li manuscript uses the same logic but applies it to a narrower claim: whether the selective electromagnetic-transfer candidate survives the registered abundance, CMB, spectral and nuclear-rate gates.

Table 1a. External references as analytical gates and comparators

External reference	Link	Role in the argument	Effect on the current claim
Burns 2026 Physics Reports / BBN sensitivity atlas	Open source	Publication-style comparator: separates observations, theory framework, sensitivity coefficients, uncertainty budgets and illustrative lithium application.	The BBN-Li paper should mirror this logic: each external reference defines a gate or comparison, not merely a bibliography entry.
arXiv:2603.22414	Open source	Open version of the Burns BBN sensitivity review and table structure.	Provides a direct model for linking reaction sensitivities and uncertainty budgets to the lithium problem.
BBN sensitivity atlas GitHub	Open source	Public numerical atlas with response functions, uncertainty budgets and PRIMAT/NACRE-II rate-compilation comparisons.	Used as the main external response benchmark for the September 12 atlas import and NACRE-II stress branch.
Fields 2011 lithium review	Open source	Problem-definition reference for the primordial lithium anomaly.	Frames why a successful mechanism must reduce mass-7 without breaking the otherwise successful BBN light-element pattern.
Cybur Fields Olive Yeh 2016	Open source	Precision standard-BBN reference after CMB-era baryon-density constraints.	Defines why broad expansion or baryon-density retuning is a high risk route and why D/H and helium gates are non-negotiable.
Pitrou Coc Uzan Vangioni 2018 PRIMAT	Open source	Precision BBN and PRIMAT reference, especially for helium and weak-rate treatment.	Supports keeping PRIMAT cross network comparison separate from a purely local LINX response result.
LINX code paper	Open source	Differentiable BBN network and joint CMB/BBN computational route.	Supports the local response derivative/eigenmode-validation strategy, while still requiring direct mediator-aware propagation.
LINX arXiv preprint	Open source	Open preprint and implementation description for Light Isotope Nucleosynthesis with JAX.	Supports the technical decision to use LINX for fast response scans and direct finite-pull reruns.
ACROPOLIS electromagnetic cascade framework	Open source	External cascade-physics precedent for electromagnetic decays and light-element photodisintegration.	Defines why a late electromagnetic-transfer candidate needs cascade and threshold gates, not just a lithium-yield fit.
Planck 2018 cosmological parameters	Open source	CMB baseline for baryon density, cosmological parameters and compressed-background consistency.	Motivates replacing compressed CMB priors with the native PR4/NPIPE nuisance posterior used in September.
COBE FIRAS CMB spectrum	Open source	Spectral-distortion observational reference.	Defines the need for spectral safety gates for any post-BBN electromagnetic energy release.
Khatri and Sunyaev 2012 CMB spectrum formation	Open source	Thermalisation and blackbody photosphere physics reference.	Frames why mu-distortion and thermalisation checks are part of the candidate safety case.
CosmoVerse White Paper 2025	DOI	Broad review of H0, BAO, CMB, BBN, systematics, varying constants, dark-sector extensions and parity-sensitive early-Universe physics.	Added as an external gate reference: lithium improvement must survive D/H, Yp, He3/D, CMB, spectral, nuclear-rate and cross-dataset checks before promotion.
Wang et al. 2024 GALAH 3DNLTE lithium catalogue	Open source	Large stellar-lithium catalogue used to build the empirical censored plateau/depletion model.	Defines the revised empirical stellar-lithium target and its limits.
Launders, Giovanetti and Liu 2026 deuterium GP analysis	Open source	Highest-priority source for synchronised, correlated D-chain functional uncertainty.	Defines the current external nuclear-rate promotion gate.
Pisanti et al. 2021, Primordial Deuterium after LUNA	Open source	Published coefficient/error-budget basis used for the executable partial external covariance gate.	Defines the current external nuclear-rate promotion gate.
Mossa et al. 2020 LUNA deuterium-burning measurement	Open source	Experimental $d(p,\gamma)^3\text{He}$ anchor underlying the improved deuterium-rate programme.	Defines the current external nuclear-rate promotion gate.
Gomez Inesta, Iliadis and Coc 2017 Bayesian d+d rates	Open source	Bayesian treatment of the paired $d(d,n)^3\text{He}$ and $d(d,p)t$ reaction rates.	Defines the current external nuclear-rate promotion gate.
Moscoso et al. 2021 Bayesian $d(p,\gamma)^3\text{He}$ rate	Open source	Independent hierarchical Bayesian rate analysis for the third Dsensitive reaction.	Defines the current external nuclear-rate promotion gate.

This reference-led structure changes the burden of proof. A lithium-only improvement is not sufficient.

The candidate must also remain compatible with D/H, Yp, He3/D, the CMB baryon-density posterior, spectral-

distortion limits, cascade physics and nuclear-rate uncertainty.

Where an external reference is weaker, incomplete or compilation-dependent, the manuscript keeps that point as an open gate rather than promoting the model.

The CosmoVerse 2025 white paper is now included as a broad external gate reference. It does not treat lithium, H0, BAO, CMB, growth, varying constants or calibration as isolated problems.

For this BBN-Li candidate the August lithium cell must therefore be judged against D/H, Yp, He3/D, Planck/ACT/SPT CMB consistency, spectral safety, BAO+BBN calibration and local-systematics controls.

The September 16 update adds an explicit CMB blackbody validation gate.

This gate asks whether the same electromagnetic-transfer idea that may reduce mass-7 can remain compatible with the near-perfect FIRAS CMB spectrum, photon-number accounting, entropy conservation and CMB temperature-redshift tests.

The work sits inside a long literature on the lithium anomaly, precision BBN, stellar depletion, electromagnetic cascades, CMB constraints and spectral distortions.

The references listed below are used as the current interpretive frame.

They define the observational problem, the standard BBN baseline, the cascade machinery and the CMB/spectral gates.

Table 1. Literature cross-reference

Topic	Representative references	Role in this paper
Problem definition	Fields 2011; Cyburt, Fields and Olive 2008; Spite and Spite 1982; Sbordone et al. 2010; Ryan 2023.	Defines the Li-7 excess relative to metal-poor stellar plateaux.
Precision BBN	Cyburt et al. 2016; Pitrou et al. 2018; Fields et al. 2020; LINX 2024.	Defines standard abundance calculations and CMB-compatible baryon density handling.
Observational anchors	Cooke et al.; helium determinations; Planck Collaboration 2020.	Defines D/H, Y _p and CMB posterior constraints.
Cascade and late-decay physics	Depta et al. ACROPOLIS 2021 and related electromagnetic-decay constraints.	Tests whether a late electromagnetic process damages other light elements.
CMB and spectral gates	Planck PR4/NPIPE/CamSpec; Fixsen FIRAS; Khatri and Sunyaev.	Tests CMB consistency and spectraldistortion safety.

Abstract

A staged investigation of the cosmological lithium problem is reported in a reference-integrated form.

July 2026 broad BBN and cosmological controls produced no joint D/H, helium and Li-7 solution; August identified a selective post-BBN electromagnetic-transfer candidate.

September Planck PR4/NPIPE, CMB-blackbody and GALAH analyses retained a constrained observational target, while an external D-chain covariance audit rejected ordinary quoted rate uncertainty as the required explanation under the registered mapping.

The present update completes a 10,000-draw fluence-matched NLO-proxy run, convergence-controlled direct LINX checks, an action-conditioned lifetime audit, an exact ACROPOLIS scan, modern-abundance and cosmology replay, a parent-decay source bridge and independent cascade tests.

The exact ACROPOLIS scan evaluated 15,768 rows and found 58 all-D/H passes. The best conditional row has m_{phi} = 4.4 MeV, tau = 150,000 s, Li7/H = 1.238279 x 10⁻¹⁰ and abundance chi2 = 0.904474.

The minimal inverse-decay action supplies at most 45.3% of the required scalar abundance, so it fails the production gate. A calibrated parent-decay bridge passes its staging gates and gives $\text{Li7/H} = 1.291358 \times 10^{-10}$, but its branching ratio is not independently predicted.

An independent reduced cascade reproduces the selected action-history candidate to 0.318%, yet only 40/58 classifications agree over the full basin. Explicit Breit-Wheeler and active-band redistribution calculations recover the targeted boundaries to within 1.588% of ACROPOLIS, with one paired control unresolved.

That control, basin_052, lies at GALAH empirical CDF 0.02403; its ESS-aware interval spans the preregistered 0.025 boundary and a 0.104% Be7 cross-section shift changes the classification.

The paper therefore reports a conditional exact-cascade parameter basin and a progressively narrowed source mechanism, not a solution. Scientific promotion remains gated by a predictive production action, exact external D-chain and mass-7 posterior products, an external full coupled cascade or independent nuclear response, and complete FIRAS/ACT/SPT closure.

1. Introduction

The cosmological lithium problem is unusual because standard BBN is not globally broken.

Precision-BBN references and CMB-era reviews show that D/H and helium largely support the standard thermal history, while lithium-7 remains high relative to metal-poor stellar plateau observations.

The external literature therefore imposes the central constraint on this paper: a viable candidate must reduce the mass-7 reservoir without using a broad background retuning that spoils deuterium, helium, helium-3 or the CMB baryon-density posterior.

This paper records the transition from negative July controls, through the positive August candidate and September CMB, stellar and covariance qualification, to the 28 September-2 October source-action and cascade programme. The chronology matters: each later calculation tests a previously open physical link and preserves both successful and failed gates.

2. Methods

The computational programme used LINX and PRIMAT for BBN network and abundance validation, ACROPOLIS 1.3.1 for exact electromagnetic decay cascades, Cobaya and CAMB for Planck likelihood sampling, and project-specific source-history, production, transport and covariance audits. The October sequence also implemented independent reduced Compton-recycling, angle-integrated Breit-Wheeler and photon-photon redistribution calculations to localise disagreements with ACROPOLIS.

External references determine the interpretation: LINX and PRIMAT define network controls; ACROPOLIS defines the full-cascade benchmark; Planck and FIRAS define cosmological and spectral gates; GALAH defines the current stellar-surface lithium likelihood; and Pisanti/LUNA, LGLL, TUNL, BRICK and de Souza define the independent nuclear-uncertainty products required for promotion.

Table 2. Method blocks

Method block	Implementation	Decision role
Standard and modified BBN controls	July LINX/FR/SU2 scripts.	Negative control and failure-boundary definition.
Selective transfer/cascade checks	ACROPOLIS plus LINX candidate pipeline.	Candidate discovery and abundance safety screen.
Finite-temperature production	Momentum-resolved and resummed production calculations.	Tests whether the candidate abundance is physically plausible.
Native CMB nuisance likelihood	Cobaya/CAMB with Planck PR4/NPIPE CamSpec, low-I TT, low-I EE and PR4 lensing; split-chain, autocorrelation-ESS and registered-cell density audit.	Replaces the compressed CMB gate and determines whether the native likelihood is technically qualified.

3. Results

3.1 July Negative Controls

The July programme reproduced the anomaly and tested whether broad modifications could resolve it.

The result was negative: no tested global modification simultaneously preserved deuterium, helium and lithium.

This result is scientifically useful because it sharply constrains what a successful mechanism can look like.

TV index

Traceability item	Recorded value
Stage	July 16-18, 2026.
Result	No D/H + helium + Li-7 pass from standard, FR/clock-proxy, reaction-rate or SU2-style control scans.
Consequence	Future solution must be selective rather than a broad background retuning.

3.2 August Selective Candidate

The August candidate uses a narrow electromagnetic window after ordinary BBN.

Its central physical premise is that mass-7, mostly Be-7 before electron capture, can be reduced without crossing into broad deuterium destruction.

The candidate therefore targets the unusual topology of the mass-7 pathway rather than changing the whole thermal history.

Table 3. Registered August candidate

Quantity	Value
mediator_mass_MeV	4.44
photon_energy_MeV	2.22
lifetime_s	1.584893e+05
n_phi_over_n_gamma_at_10MeV	1.211528e-06
g_phi_gamma_GeV_inv	9.767274e-11
D_H_times_1e5	2.493161
Yp	0.246735
Li7_H_times_1e10	1.488554
He3_D	0.415242
eta10_CMB	6.096517
Neff_CMB	3.044601
mu_distortion	1.476434e-09
joint_pass_probability	0.650158
delta_log_evidence_vs_standard	105.206012
Planck bestfit eta10	6.105374
Planck bestfit nnu	2.84265
Planck bestfit YHe	0.282034
Planck bestfit chi2_CMB	10971.285

TV index

Traceability item	Recorded value
Stage	August 20-21, 2026.
Positive result	Candidate passes the recorded abundance/cascade/FIRAS gates at the frozen cell.
Boundary	A complete action-level particle model is outstanding.

3.3 Native Planck CMB Nuisance Qualification

The native Planck PR4/NPIPE test is the strongest new constraint in this update.

It lets ω_b , N_{eff} and Y_p vary independently, and marginalises over foreground and calibration nuisance directions rather than using only a compressed two-variable Planck prior.

Table 4. Qualification gates

Gate	Threshold	Observed	Pass
Cobaya split-chain mean R-1	≤ 0.03	0.0293	True
Independent split R-1 max over ombh2, nnu, YHe, nuisance	≤ 0.03	0.0021	True
Autocorrelation ESS min over ombh2, nnu, YHe, nuisance	≥ 1000	1449.0004	True
Registered lithium cell joint posterior density	$\Delta\chi^2 \leq 7.814728$	1.2574	True
Nuisance split stability diagnostic	max split mean shift ≤ 1 sigma	0.0838	True
First-half vs second-half stability diagnostic	max mean shift ≤ 1 sigma	0.0926	True

Table 5. Registered-cell posterior density

Quantity	Value
posterior mean η_{10}	6.074645
posterior mean N_{eff}	2.814036
posterior mean Y_p	0.262997
registered η_{10}	6.096365
registered N_{eff}	3.044709
registered Y_p	0.246733
registered $\Delta\chi^2$	1.25743
Mahalanobis distance	1.121352
density ratio to posterior peak	0.533277
posterior mass inside equal-density contour	0.261374
joint 95 gate pass	True
weighted kNN density ratio	0.601247

Table 6. CMB posterior summary

Parameter	Mean	Std	Median	68% interval	95% interval
ombh2	0.02219	1.9507e-04	0.02219	0.021995 to 0.022381	0.021809 to 0.022573
nnu	2.81855	0.2532	2.81194	2.569554 to 3.062147	2.343549 to 3.327882
YHe	0.262427	0.0186	0.262668	0.244057 to 0.280933	0.225078 to 0.298364
η_{10}	6.075084	0.0534	6.075231	6.021877 to 6.127459	5.97098 to 6.179927
H0	66.210921	1.5058	66.164438	64.706901 to 67.68329	63.421067 to 69.23348
omegam	0.31734	0.0087	0.317346	0.308729 to 0.326008	0.300514 to 0.334223
σ_8	0.803256	0.0095	0.803151	0.793847 to 0.812604	0.784934 to 0.822145
S8	0.826006	0.0111	0.826026	0.814952 to 0.837015	0.804369 to 0.847849
χ^2_{CMB}	10984.919918	7.9961	10984.146	10979.26564 to 10990.31448	10975.553 to 10997.62775

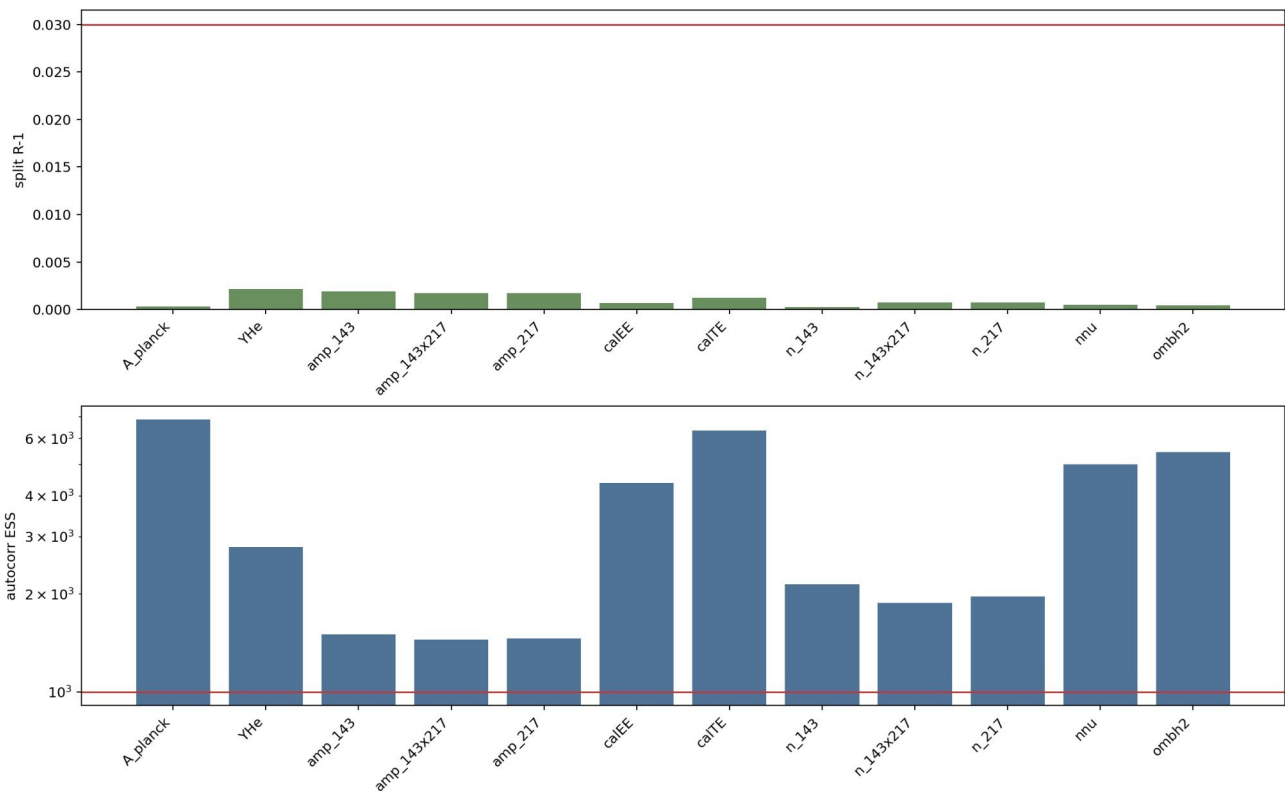


Figure 1. Combined main plus ESS top-up qualification gates. TV index

Traceability item	Recorded value
Stage	September 11, 2026.
Combined chain size	120073 compressed rows; raw weight sum 393272.
Gate outcome	PASS on all preregistered gates.
Most conservative ESS	1449 for amp_143x217 among audited gate parameters.

3.4 September 2026 NLO and Anchor-Sensitivity Updates

After the combined Planck PR4/NPIPE qualification audit, three September checks were added to test whether the August candidate survives extended sources of modelling uncertainty: a matched NLO thermal gate, a generic abundance-anchor sensitivity grid, and a named literature-anchor stress test.

These checks are not a new discovery claim; they are designed to state exactly which abundance assumptions leave the candidate viable.

- Matched NLO thermal gate:** 625/625 pass.
 Weighted NLO shift stayed within +/-2.569%; maximum single-cell total shift 2.741%; pole-cut consistency and thermal self-energy proxies remained inside the declared gate.
- Generic anchor sensitivity:** 486/1152 full pass; 15/18 registered-anchor pass.
 Failures were concentrated in the low-D/H plus high deuterium-destruction direction, showing that the candidate is sensitive to the deuterium anchor rather than freely adjustable.
- Named literature-anchor stress test:** 1020/2160 full pass; 15/18 registered literature/project cells pass.
 The registered cell remains viable, while high single-system D/H, EMPRESS-low helium and Izotov-high helium stress choices fail in this simplified gate.

The practical interpretation is narrower but stronger: the August candidate does not disappear under the September gate checks, but it should be written as conditional on the declared D/H, helium and nuclear-rate assumptions until the same point is reproduced inside an independent BBN solver with full nuclear covariance propagation.

3.5 September 12 Nuclear Rate and Cross Network Validation

After the September 11 paper draft, the nuclear-rate branch was expanded to test whether the registered lithium candidate is limited by software response, cross-network disagreement, observational anchors or the absence of a true reaction-by-reaction covariance source.

These checks are ordinary-BBN rate-pull diagnostics and response-propagation tests; they do not yet inject a physical mediator directly into PRIMAT, ParthENoPE or an independent network.

Table 7. September 12 nuclear-rate and direct-network validation summary

Check	Quantitative result	Interpretation
PRIMAT N=5000 rate Monte Carlo bridge	5000 samples; 12/12 mapped reactions; He3/D propagated; pass fraction 0.6036; median abundance χ^2 4.6314.	Covariance sampling is stable enough for a diagnostic, but the registered abundance gate remains incomplete.
N=1000 versus N=5000 covariance stability	Pass fraction moved from 0.6116 to 0.6036; Frobenius covariance drift 0.1158; dominant failures through D/H.	Increasing sample size did not rescue the gate. D/H is the limiting abundance channel.
Direct LINX eigenmode validation on N=5000 covariance basis	32/32 direct LINX runs completed; maximum D/Yp/Li response error 0.01023 observational-sigma units; He3/D max absolute error 0.0001677.	The local first-order response approximation remains technically validated along the tested covariance eigenmodes.
BBN sensitivity-atlas and NACRE-II stress branches	Five million draws per scenario; PRIMATsource independent pass fractions near 0.608; NACRE-II independent pass fractions near 0.170.	PRIMAT-like responses align with the local LINX bridge; NACRE-II behaves as a strong rate-compilation stress branch.
Direct PRIMAT abundance reruns against reused exact LINX points	51/51 matched points completed; maximum direct D/Yp/Li disagreement 0.01137 observational-sigma units.	Cross-network disagreement is not the dominant failure mode at the tested ordinary-BBN rate-pull points.
Fresh exact LINX D-chain direct reruns	31/31 direct points completed; worst disagreement 0.02673 observational-sigma units at D_chain_all_positive_3, dominated by Li7/H; D/H grid pass fraction 0.54996.	Even the deliberately stressed D-chain direct points agree well between LINX and PRIMAT; the open gate is covariance and mediator physics, not direct-network mismatch.

The main inference is therefore narrower and cleaner than before.

The direct-network and response-linearity checks support the computational bridge, but the rate-covariance gate is still not closed because a large share of rate draws miss the registered joint abundance cell, especially through D/H. The most important failure-associated reaction directions remain dpHe3g, ddHe3n and ddtP.

This section should be read as a qualification update. It strengthens the evidence that the September BBN-Li programme is internally reproducible across LINX and PRIMAT at the tested points, while preserving the requirement for a published or package-native nuclear-rate covariance input and a direct physical mediator propagation test.

3.6 Reference Led Claim Gates

The external literature can be read as a set of claim gates. Problem-definition papers require a mass-7 reduction large enough to reach the observed lithium plateau.

Precision-BBN and CMB papers require D/H, Yp, eta10 and N_eff to stay inside their accepted ranges.

Cascade and spectral-distortion papers require late electromagnetic energy transfer to avoid deuterium destruction and blackbody-spectrum violations.

Sensitivity-atlas and rate-compilation papers require the result to survive reaction-rate uncertainty and PRIMAT/NACRE-II comparison.

Gate	External basis	Current BBN-Li status	Next required action
Lithium anomaly	GALAH DR3 3D-NLTE with Gaia controls; locked DR4 release validation; Fields/Spite context.	Target sharpened: $A(\text{Li}) = 2.09062$ [2.03076, 2.15345]; 3/46 earlier candidates survive the qualified posterior gate.	Replicate with an independent stellar survey and a physical stellar-depletion calculation.
D/H and helium preservation	Direct LINX candidate propagation plus the registered D/H, Yp and He3/D anchors.	Conditionally satisfied in 58 exact ACROPOLIS rows and in the action-history candidate; modern PRIMAT replay gives up to 11.4% all-D/H pass.	Repeat with a predictive source and exact synchronised external D-chain posterior draws.
CMB baryon-density compatibility	Planck PR4/NPIPE nuisance posterior, compared against Planck-era cosmology.	Qualified under the combined main plus ESS top-up audit.	Extend to ACT/SPT where likelihood consistency allows.
Cascade and spectral safety	ACROPOLIS, FIRAS and Khatri/Sunyaev thermalisation physics.	Exact ACROPOLIS basin found; candidate 17 independently reproduced to 0.318%; full reduced basin and one active-band control remain open.	External full coupled cascade or independent nuclear response, then full FIRAS covariance.
Nuclear-rate uncertainty	Pisanti/LUNA partial coefficient covariance; LGLL GP and TUNL Bayesian posterior targets.	Failed for ordinary uncertainty under the current mapping: 0/60 inside the available external 95% region.	Obtain exact synchronised LGLL draws or paired TUNL posterior samples; do not substitute the internal proxy.

This is the reason the present claim remains deliberately narrow.

The result is stronger than a lithium curve fit because it is cross-checked against external observational and computational structures.

It is still not a final solution because the external structures also identify the remaining failure modes.

3.7 CosmoVerse Tension Review Gate

The CosmoVerse review gives a useful check on the interpretation of the September BBN-Li programme.

It supports treating the lithium anomaly as real and persistent, but it also keeps stellar depletion, observational systematics and nuclear-rate uncertainty in play.

The present result therefore remains a candidate mechanism under test rather than a claimed resolution.

For the wider PLAMB and SU2R programme, the same review acts as a warning against single-dataset promotion.

A raw-MU or distance-law improvement must survive BAO geometry and CMB recombination constraints, while an SU2R background improvement must survive a perturbation and high-resolution CMB closure path.

Those requirements are now treated as linked gates rather than optional follow-up work.

Programme branch	CosmoVerse issue	Current interpretation	Next gate
BBN-Li	Lithium, D/H, Yp, He3/D and nuclear-rate uncertainty must be considered jointly.	The August cell remains promising, but September covariance and direct-network work keep the scientific gate incomplete.	Physical mediator-aware abundance propagation and a published or package-native covariance source.
PLAMB/FR	H0, BAO, CMB and local distance-ladder systematics are cross-coupled.	Raw-MU linearity is not enough by itself; no-expansion claims must pass standard ruler and recombination tests.	BAO+BBN consistency, distance-duality/opacity checks, redshift-frame and local-flow audits.
SU2/SU2R	Dark-sector extensions need growth, perturbation and CMB consistency, not only background-distance fits.	Effective-fluid status is retained; canonical nonphantom SU2R collapses toward LCDM in the current DESI-era run.	Freeze action-level route before direct ACT/SPT or perturbation-likelihood promotion.
Chiral diagnostics	Parity violation, primordial magnetic fields and chiral gravitational waves are relevant early-Universe probes.	These are useful diagnostics, but not direct late-time likelihood evidence for BBN-Li or SU2R.	Parity-observable likelihoods and CMB EB/TB or four-pointfunction hooks.

This gate improves the paper's publication structure because it ties the candidate mechanism to the wider observational-tension landscape.

It also preserves the negative result boundary: the BBN-Li candidate has not yet cleared the final covariance or mediator-aware propagation gates.

3.8 CMB Blackbody Validation Gate

The BBN-Li candidate must now pass an independent CMB spectral-shape gate.

A mechanism that selectively changes the late electromagnetic energy budget can reduce lithium in an abundance table but still fail cosmology if it distorts the CMB away from a Planck blackbody, changes photon number after light-element freeze-out, or shifts the baryon-to-photon ratio used by the CMB posterior.

The September 16 FR/CMB diagnostic is deliberately conservative.

It is not an official FIRAS covariance likelihood.

It tests representative Planck-preserving, grey, non-grey, dispersive and temperature-mixture transport classes on a 30-3000 GHz grid and refits the best grey amplitude and blackbody temperature before scoring residuals. The result is a useful gate-setting scale: blackbody-preserving transport remains viable at the shape level, while non-grey and dispersive variants become constrained quickly.

Validation test	What it checks	Implication for BBN-Li
Thermalisation epoch classification	Classify whether mediator electromagnetic transfer occurs before the blackbody thermalisation surface, during the μ/γ distortion era, or after the CMB spectrum is effectively frozen.	A pre-thermalisation transfer is safer; a late transfer must predict sub-FIRAS nongrey, μ/γ and temperature-mixture residuals.
FIRAS-style spectral-shape projection	Project any photon-sector transfer into Planck-curve residuals after a best grey amplitude and temperature refit.	The current 50 ppm RMS proxy gives a quick exclusion scale and must be replaced by an official FIRAS covariance likelihood before promotion.
Photon number and entropy accounting	Track whether the proposed lithium-sector transfer changes photon number, photon energy density, or η_B/γ after light-element freeze-out.	D/H, Yp and the CMB baryon-density posterior are sensitive to entropy or photon dilution, so lithium-only tuning is insufficient.
Direct mediator-aware abundance propagation	Run the mediator model through LINX and an independent network rather than applying a response-level abundance shift.	This remains the main open scientific gate: current direct LINX/PRIMAT checks validate ordinary rate-pull plumbing, not the mediator dynamics.
CMB temperature-redshift consistency	Check whether any FR or photon transport law preserves the Planck occupation structure and the observed $T(z)$ relation.	A model can preserve blackbody shape only if $h\nu/kT$ is transported coherently; non-grey or dispersive variants are tightly bounded.

The current 50 ppm RMS proxy limits, after best grey-amplitude and temperature refit, are:

Distortion family	Parameter	Approximate allowed amplitude	BBN-Li interpretation
Non-grey photon-transfer tilt	epsilon_tilt	3.27067597e-05	A frequency-dependent brightness or clock-rate tilt must be below this proxy amplitude after best blackbody refit.
Dispersive redshift remap	epsilon_dispersion	4.43003370e-07	A frequency-dependent redshift remapping is very tightly constrained because it warps the Planck curve.
Line-of-sight temperature mixture	sigma_InT	4.02746473e-04	Path-dependent thermal or clock scatter cannot superpose a broad distribution of blackbodies without visible residuals.
Chemical-potential distortion	mu	1.43389573e-04	Standard mu-type spectral distortion reference scale for energy release before full late thermalisation is lost.
Compton y distortion	y	8.48604116e-08	Standard y-type reference scale for late Comptonisation like distortion in the same proxy gate.

Interpretation: this gate does not reject the August candidate cell, because the candidate has not yet been propagated as a fully specified photon-transfer history. It does, however, narrow the safe theory space.

A successful mechanism should either thermalise early enough to preserve the Planck occupation structure, or produce residual mu, y, non-grey and temperature-mixture amplitudes below FIRAS-level limits.

It should also leave D/H, Yp, He3/D and the CMB baryon-density posterior mutually consistent.

Linked program outputs: [diagnose_fr_cmb_blackbody_transform.py](#); [diagnose_fr_cmb_distortion_limits.py](#); summary outputs under plamb_runs/diagnostics/fr_cmb_blackbody_transform and plamb_runs/diagnostics/fr_cmb_distortion_limits.

3.9 GALAH Recalibration, Direct LINX Propagation and External D-Chain Covariance

The September 23-24 programme applies three distinct evidential layers.

The GALAH analysis changes the observational lithium likelihood; the adaptive search and direct LINX batch test whether abundance-compatible candidate rows exist; the external covariance audit asks whether those rows are compatible with independently reported nuclear-rate uncertainty.

These layers must not be collapsed into one pass fraction.

Table 8. September 23-24 evidence chronology

Stage	Design	Quantitative result	Interpretation boundary
23 Sep: GALAH DR3 model	977 stars; 734 detections and 243 upper limits; Gaia-informed censored mixture model.	$A(\text{Li}) = 2.09062$, 95% interval 2.03076-2.15345; max split R-hat 1.06886; min ESS 1334.7.	Technical pass; empirical stellar surface anchor, not a primordial abundance determination.
23 Sep: locked GALAH DR4 check	No coefficient, prior, cut or imputation value refitted after opening DR4.	168 strict Gaia-source non-overlap stars; detection bias +0.0502 dex.	Useful release check, but not independent-survey replication.
23 Sep: archive re-score	277 CSVs audited; 70 compatible files; 244,867 complete, deduplicated headline rows.	GALAH-posterior passes 21,737 versus 89,285 under the old fixed anchor; 24.35% retained.	Reclassification result; it does not rerun network dynamics.
23 Sep: adaptive shotgun	200,000 hypothesis-search rows using mediator and D-chain response structure.	138,797 posterior-gate passes; 56,431 pure-template passes; 5,529 within the tested internal q envelope.	Search enrichment only; pass fractions are not evidence probabilities.
24 Sep: direct LINX propagation	60 selected mediator plus D-chain rows propagated in LINX.	60/60 technical completions and 60/60 active GALAH abundance gate passes; best $\chi^2 = 1.83743$.	Direct in LINX, but still an operator split source prototype and not an independent-network mediator calculation.
24 Sep: external covariance audit	The same 60 rows were tested against the available partial Pisanti/LUNA three-reaction covariance.	0/60 inside the 95% region over $T_9 = 0.1$ -1.5; best $D_M^2 = 136.914$ versus threshold 7.815.	Strong negative result for ordinary quoted D-chain uncertainty under the registered mapping.

The direct 60-row queue was selected from a search designed to find successful abundance combinations. Consequently, 60/60 direct abundance passes demonstrate reproducibility of those selected candidates in LINX; they do not estimate a prior predictive probability or a discovery significance.

Table 9. Best abundance-passing candidate under the external covariance audit

Quantity	Value
Candidate	combined_gaussian_T32_g2p25_s0p65__q_-1p5712_-2p3568_-2p3568
Direct abundance outputs	$D/H = 2.530419\text{e-}5$; $Y_p = 0.247566$; $\text{Li7/H} = 1.393770\text{e-}10$; $\text{He3/D} = 0.404709$
Abundance χ^2	1.907920
Most favourable tested temperature	$T_9 = 0.4$
External Mahalanobis statistic	$D_M^2 = 136.913980$; $D_M = 11.701025$
External 95% threshold	$D_M^2 \leq 7.814728$ for three effective rate dimensions
Tail probability	1.74908e-29
Classification	ABUNDANCE_PASS_EXTERNAL_COVARIANCE_FAIL

The rate-pull coordinate q_i was converted to a physical fractional rate displacement using the PRIMAT temperature-dependent factor uncertainty $f_i(T_9)$:

$$\delta_i(T_9) = \exp[q_i \ln f_i(T_9)] - 1 \quad (44)$$

Compatibility with the external fractional-rate covariance C was then scored by

$$D_M^2 = \delta^T C^+ \delta \quad (45)$$

where C^+ is the Moore-Penrose inverse.

The audit evaluated $T_9 = 0.1, 0.2, 0.4, 0.8, 1.0$ and 1.5 and reported the smallest distance, deliberately giving each candidate its most favourable tested mapping.

Table 10. Best candidate D-chain displacements at $T_9 = 0.4$

Reaction	Code label	Required shift	External 1-sigma	Marginal z
$d(p,\gamma)3\text{He}$	dpHe3g	-3.2126%	1.2146%	-2.645
$d(d,n)3\text{He}$	ddHe3n	-2.5454%	0.3714%	-6.853
$d(d,p)t$	ddtp	-2.5454%	0.2795%	-9.108

The d+d branches dominate the incompatibility. Their required shifts are only a few per cent in absolute terms, but the available external marginal uncertainties are below four-tenths of one per cent. Subsequently a modest rate displacement results in a large covariance distance.

Table 11. External D-chain posterior source hierarchy

Priority	Source	Current status	Required action
1	LGLL 2026 Gaussian-process posterior	Exact synchronised function/rate draws not yet released locally.	Request common-draw S-factor functions or thermally averaged rate draws on a shared T9 grid.
2	TUNL Bayesian D-rate posteriors	Published marginal analyses located; paired samples are not available locally.	Obtain paired d+d samples and a compatible d(p,gamma)3He chain.
3	Pisanti/LUNA coefficient covariance	Available and tested as independent per-reaction blocks; cross-reaction covariance is absent.	Retain as a partial external gate, not a complete joint posterior.
4	ETR25/RateMC archive	Archive checksum verified; no applicable D-chain files found.	Use for marginal-rate validation only if matching reaction products become available.
5	IAEA EXFOR reconstruction	Not yet built.	Preregister a joint experimental likelihood only if posterior products cannot be obtained.

Scientific interpretation: the present external test is partial because it combines independent per-reaction coefficient-covariance blocks and lacks released cross-reaction and shared-normalisation covariance.

Even with that favourable limitation, the best candidate is far outside the 95% region.

The result strongly rejects an ordinary-error explanation under this mapping.

It is not a no-go theorem for a new interaction that physically modifies effective reaction flow; that alternative must be judged from its action and source dynamics, not from the standard-rate covariance.

TV index

TV item	Assessment for the September 23-24 update
TV1-TV3: problem, observable, model	Observable gates are explicit; the source remains phenomenological rather than action-derived.
TV4-TV6: assumptions, mathematics, computation	Rate mapping, covariance statistic, temperature grid and direct LINX path are reproducible.
TV7-TV9: uncertainty, controls, falsification	The external covariance is a genuine falsifying control and returns a clear negative result for ordinary rate uncertainty.
TV10-TV12: comparison, novelty, promotion	Novelty is the linked observational-anchor/direct-propagation/external-covariance chain; promotion remains blocked.

4. October 2026 Action-Derived Source and Cascade Qualification

The 28 September to 2 October programme replaced the earlier response-level source with progressively more physical tests. The sequence was deliberately ordered: fluence calibration, direct LINX discretisation checks, a minimal action-derived lifetime and shut-off law, exact ACROPOLIS decay cascades, production and cosmology screens, and independent reduced-cascade reproduction. Each pass and failure is retained because they constrain different parts of the proposed mechanism.

4.1 Full-Fluence and Direct-Network Qualification

The fluence-matched NLO-proxy PRIMAT run completed 10,000/10,000 draws. It retained 1,785 precision-nine passes and 1,433 rows that passed every registered D/H branch. These fractions describe a hypothesis-search distribution under the declared partial covariance; they are not a posterior probability for the mediator model.

A stratified direct LINX rerun then completed four convergence-controlled rows, including the source-off control. Three rows passed the technical validation criterion, the largest abundance difference was 0.325 sigma, and two of the three source rows retained the same precision-nine and all-D/H classifications. The low-temperature-tail audit subsequently showed that the best row was numerically converged but source-prescription sensitive: 82.0% of its baseline exposure came from the low-temperature tail, and neither floor removal nor cutoff robustness passed. This converted an apparent numerical success into a physical-source question.

4.2 Action, Production and Exact ACROPOLIS

The minimal scalar action used $V(\phi) = m_\phi^2 \phi^2 / 2$ and $L_{\phi\text{-}\gamma} = -g_{\phi\text{-}\gamma} \phi F_{mn} F^{mn} / 4$. Expansion dilution and the lifetime were derived from this ansatz, while the nuclear response normalisation remained calibrated to the frozen LINX endpoint. None of the 126 primary s-wave, p-wave or mild inverse-speed rows passed; only diagnostic responses rising approximately as T^β with beta no larger than -3 produced passes. The minimal EFT source therefore failed and a resonance, threshold enhancement or additional production channel became necessary.

The native ACROPOLIS 1.3.1 scan evaluated 15,768 decay-mediator rows without a runtime failure. Fifty-eight rows passed the precision-nine and all-D/H gates. The best conditional row had $m_\phi = 4.4$ MeV, $E_\gamma = 2.2$ MeV, $\tau = 150,000$ s, $n_\phi/n_\gamma = 1.53993 \times 10^{-6}$, $D/H = 2.512784 \times 10^{-5}$, $Y_p = 0.2469982$, $Li7/H = 1.238279 \times 10^{-10}$ and abundance $\chi^2 = 0.904474$. This is an exact cascade and photodisintegration result for a scanned initial scalar abundance, not a derivation of that abundance.

When the 58 rows were replayed against 10,000 correlated modern PRIMAT abundance draws, every row retained at least one abundance pass; the largest precision-nine and all-D/H pass fractions were 27.95% and 11.40%. The eta, N_{eff} and FIRAS visibility screens were largely benign, but inverse-decay freeze-in supplied at most 45.3% of the required scalar population. No row passed the combined production screen. A separately declared parent-decay source then produced a candidate-17 bridge that passed all eight registered staging gates, with $Li7/H = 1.291358 \times 10^{-10}$ and $\chi^2 = 1.305824$, but its reheating branching ratio was calibrated rather than independently predicted.

Table 12. September 28 to October 2 qualification chronology

Stage	Quantitative result	Gate status	Scientific meaning
10,000-draw full-fluence NLO proxy	1,785 precision-nine; 1,433 all-D/H passes	Hypothesis-search complete	Large candidate basin under the declared proxy; not a model posterior.
Stratified direct LINX	4/4 completed; 3 technical passes; max 0.325 sigma	Technical partial pass	Discretisation is controlled for the tested rows, but classifications are not universal.
Low-temperature tail	82.0% baseline tail exposure; cutoff and floor-removal gates fail	Gate incomplete	The source history, rather than solver convergence, controls the candidate.
Minimal action-derived source	0/126 primary rows pass	Failed	Ordinary dilution plus minimal response does not sustain the required late source.
Exact ACROPOLIS scan	15,768 rows; 58 all-D/H passes; best $\chi^2 = 0.9045$	Conditional parameter pass	A viable exact decay-cascade basin exists when the initial scalar abundance is scanned.
Modern abundance and cosmology replay	Max all-D/H pass 11.4%; freeze-in/required ≤ 0.453	Production gate failed	Abundance and visibility can pass, but the minimal action underproduces the source.
Parent-decay action bridge	8/8 staging gates; $Li7/H = 1.2914 \times 10^{-10}$	Technical pass	Time staging closes conditionally; the parent branching remains calibrated.

Stage	Quantitative result	Gate status	Scientific meaning
Independent reduced basin	40/58 classifications agree; 54/58 within 20%; 11-row robust core	Global gate incomplete	Candidate 17 is robust, but hard pair-opening treatment fails across the full basin.
Explicit pair-production audit	11/11 core; 6/11 neighbours; all 22 within 10%	Targeted technical pass	Angle-integrated Breit-Wheeler treatment removes the hard-cutoff artefact for the core.
Nuclear-active-band redistribution	5/5 boundary rows recovered; 4/5 controls retained; max 1.588%	Technical gate incomplete	Photon-photon redistribution explains most residual discrepancy; one boundary remains.
basin_052 boundary audit	GALAH two-sided tail 0.0481; ESS interval spans the 2.5% boundary	Boundary unresolved	The lone control flip is too close to observational and cross-section boundaries for a binary claim.

4.3 Independent Cascade and Boundary Readout

The independent reduced line cascade reproduced the action-history candidate at $\text{Li7/H} = 1.287252 \times 10^{-10}$, only 0.318% from the ACROPOLIS bridge, with stable cutoff and resolution checks. Across the full 58-row ACROPOLIS pass basin, however, only 40 classifications agreed and 54 remained within 20% in lithium; 11 rows formed a diagnostic robust core. This failure was traced mainly to the hard pair-opening convention rather than numerical resolution.

Replacing that convention with the angle-integrated Breit-Wheeler rate retained all 11 robust-core classifications and 6/11 adjacent neighbours, with all 22 rows within 9.97% of ACROPOLIS. Adding repeated photon-photon redistribution in the Be7-active band recovered all five remaining boundary rows and brought all ten targeted rows within 1.588% of ACROPOLIS, but retained only four of five paired controls. The technical gate therefore remains incomplete rather than failed wholesale.

For the sole flipped control, basin_052, the primary lithium value lies at GALAH empirical CDF 0.02403, with a two-sided tail probability of 0.04805. The autocorrelation-ESS interval for that CDF is 0.01711-0.03371 and therefore crosses the registered 0.025 lower boundary. A -0.104% change in the Be7 photodisintegration scale reaches the boundary. The preferred BRICK/de Souza summary proxy places 46.27% of 500,000 draws inside the GALAH interval, reproduced at 46.20% with an independent seed. The earlier binary failure is consequently model-sensitive and unresolved, not a confirmed pass or robust rejection.

Table 13. Current promotion gates after the October cascade programme

Gate	Current evidence	Status	Required closure
Abundance target	GALAH posterior plus 10,000-draw PRIMAT replay	Qualified observational target	Independent stellar-survey and depletion replication.
Ordinary D-chain uncertainty	Best earlier candidate $D_{\text{M}}^2 = 136.914$; 0/60 in partial 95% region	Failed under registered mapping	Exact synchronised LGLL/TUNL posterior products.
Exact mediator cascade	58 ACROPOLIS all-D/H passes	Conditional parameter pass	Derive rather than scan the mediator population.
Source action	Minimal inverse decay underproduces; calibrated parent bridge passes	Incomplete	Predict parent production, branching and phase-space history from one frozen action.
Independent cascade	Candidate 17 agrees at 0.318%; active-band boundary agreement $\leq 1.588\%$	Targeted technical pass	External full coupled electron-photon cascade and independent nuclear response.
Mass-7 uncertainty	basin_052 boundary spans GALAH ESS interval; BRICK/de Souza used only as summary proxy	Unresolved	Independent Be7 photodisintegration covariance and exact synchronised mass-7 rate draws.
CMB and spectral safety	Planck qualified; η/N_{eff} and FIRAS visibility screens benign for exact pass	Screen pass, likelihood open rows	Full FIRAS covariance and ACT/SPT-consistent likelihood closure.

TV index: October evidence updates correspond to the full-fluence, action-lifetime, ACROPOLIS, production, parent-bridge, independent-cascade, explicit-pair, active-band and basin_052 registered outputs dated 28 September to 2 October 2026.

5. Discussion

The October calculations materially strengthen the candidate while also making its remaining weakness more precise. An exact ACROPOLIS decay-cascade basin exists, the best abundance row is quantitatively strong, and the action-history candidate survives a targeted independent transport calculation.

The result is not explained by ordinary quoted D-chain uncertainty, and it is not generated by the minimal inverse-decay action. The model now requires an explicit production sector or parent population whose branching, abundance and momentum history are predicted rather than calibrated.

The independent cascade programme shows why a single validation row is insufficient. Candidate 17 is stable, but the wider basin is sensitive to pair-production and photon-redistribution modelling near the Be7 threshold. Adding these processes recovers most targeted discrepancies and leaves one observationally marginal control rather than a global cascade failure.

The basin_052 audit is especially important for claim discipline. Its classification changes under sub-percent Be7 cross-section movement and its GALAH ESS interval straddles the registered lower boundary. It should therefore be reported as unresolved, not counted as either decisive support or decisive falsification.

The current hypothesis is consequently testable and narrower than the August proposal: a late electromagnetic source in the Be7-open, deuterium-closed window can work conditionally, but its population, full cascade and nuclear posterior must be closed by independent inputs.

This distinction preserves the central scientific result. The programme has progressed from a curve-fitting candidate to an exact-cascade parameter basin with identified production and boundary requirements, while stopping short of a solution claim.

6. Limitations

- The exact ACROPOLIS scan conditions on a scanned initial scalar abundance; the minimal action does not derive enough of that abundance.
- The parent-decay bridge uses a calibrated reheating branching ratio and a cohort approximation rather than a fully sampled particle-production model.
- The independent cascade is reduced rather than a second full coupled electron-photon Boltzmann package, and it reuses the published Be7 response fit.
- The basin_052 mass-7 calculation uses independent Gaussian summary blocks because exact synchronised BRICK and de Souza rate curves and an independent Be7 photodisintegration covariance are not available locally.
- The public FIRAS table is used with visibility screens because the full correlated covariance is not yet registered, and ACT/SPT likelihood closure remains open.
- The Planck analysis is spectrum-level rather than map-level, and the abundance and CMB gates are not yet sampled in one global microscopic posterior.
- The GALAH DR4 check remains a same-survey release validation; independent stellar-depletion and external survey replication are pending.
- All search-enriched pass fractions are descriptive of the tested samples and must not be interpreted as discovery significances or model probabilities.

7. Conclusions

The July investigations failed to resolve the lithium problem with broad BBN or cosmological modifications.

The August selective electromagnetic candidate survived the September Planck, blackbody and GALAH gates, while the external D-chain covariance audit rejected ordinary quoted rate uncertainty as its explanation.

The October work establishes a conditional exact ACROPOLIS basin: 58 of 15,768 rows pass every registered D/H branch, with a best abundance χ^2 of 0.904474.

The mechanism is not yet closed. Minimal inverse-decay production is insufficient; the calibrated parent bridge is technically successful but not predictive; the full independent cascade basin is incomplete; and basin_052 remains an observational and mass-7 cross-section boundary case.

The strongest defensible conclusion is therefore that a narrow late electromagnetic source remains a quantitatively viable conditional candidate with identified falsification gates. It is not yet a solution of the cosmological lithium problem.

8. Future Work

1. Freeze one parent-production action and predict the reheating yield, branching ratio, mediator phase-space distribution and late two-photon source without calibration to the abundance target.
2. Reproduce the candidate and boundary rows in an external full coupled electron-photon cascade or an independently implemented nuclear-response pipeline.
3. Obtain exact synchronised LGLL/TUNL D-chain posterior products and independent Be7 photodisintegration covariance or posterior samples; then repeat the joint gate.
4. Propagate exact BRICK and de Souza mass-7 posterior curves rather than independent Gaussian summary blocks.
5. Run full FIRAS covariance, ACT DR6 and SPT-3G likelihood checks with parameter and calibration consistency.
6. Replicate the GALAH lithium posterior in an independent stellar survey with a physical depletion model.
7. Publish a frozen reproducibility bundle with checksums for the action declaration, source history, exact cascade, independent transport, abundance inputs and final paper outputs.
8. Keep NACRE-II as a separate stress branch until reaction normalisations, response definitions and covariance assumptions are matched.

Highest-priority next tests

9. Obtain an external Be7 photodisintegration covariance or posterior sample set and exact synchronised BRICK/de Souza rate curves, then repeat basin_052 without the summary-response approximation.
10. Implement the frozen parent-decay source in a second full cascade or independent nuclear-response calculation and preregister the candidate and boundary rows.
11. Replace the calibrated parent branching with a predictive production calculation from the same action, including momentum history, inverse processes, entropy and energy accounting.
12. Obtain synchronised LGLL Gaussian-process or paired TUNL D-chain samples on a common temperature grid and repeat the external covariance gate.
13. Run official FIRAS covariance plus ACT DR6/SPT-3G likelihood closure while retaining the Planck, D/H, Yp, He3/D and GALAH gates unchanged.
14. Retain all failed and boundary rows in the public manifest so future source models are tested against the same frozen evidence set.

Appendix A. Acronyms

Table A1. Acronym glossary

Acronym	Meaning
ACROPOLIS	Public electromagnetic-cascade framework used for BBN photodisintegration constraints.
AIC	Akaike Information Criterion; lower is preferred after a penalty of 2 parameters.
BBN	Big Bang Nucleosynthesis.
BIC	Bayesian Information Criterion; lower is preferred with a stronger $\log(N)$ parameter penalty.
CAMB	Code for Anisotropies in the Microwave Background.
CMB	Cosmic microwave background.
Cobaya	Cosmological Bayesian analysis package used for the Planck nuisance MCMC.
D/H	Deuterium-to-hydrogen abundance ratio.
ESS	Effective sample size; here the autocorrelation-aware number of effectively independent posterior samples.
FIRAS	Far Infrared Absolute Spectrophotometer on COBE.
LINX	Fast differentiable BBN package used for abundance calculations and validation.
MCMC	Markov-chain Monte Carlo.
NPIPE	Planck reprocessing pipeline used for PR4 products.
PR4	Planck public release 4.
SPD	Symmetric positive definite; used for covariance handling and Cholesky/LAPACK inversion.
TV index	Traceability/validation index. The reports place it after each investigation section.
NLO	Next-to-leading order; the first correction beyond the leading thermal calculation.
EMPRESS	Extremely Metal-Poor Representatives Explored by the Subaru Survey; source of a low-helium stress anchor.
PRIMAT	Public precision BBN code used as an external comparison target for future reproduction.
ParthENoPE	Public BBN code used as an external comparison target for future reproduction.
GALAH	Galactic Archaeology with HERMES stellar spectroscopic survey.
GP	Gaussian process; a distribution over functions used here for correlated nuclear S-factor inference.
LGLL	Launders, Giovanetti and Liu 2026 deuterium Gaussian-process analysis.
LUNA	Laboratory for Underground Nuclear Astrophysics at Gran Sasso.
TUNL	Triangle Universities Nuclear Laboratory.

Appendix B. Variables

Table B1. Variable glossary

Variable	Meaning
eta10	10^{10} times the baryon-to-photon ratio η ; converted from ω_b using $\eta_{10} = 273.78 \omega_b$ for $T_{\text{CMB}} = 2.7255 \text{ K}$.
ω_b / ombh^2	Physical baryon density $\Omega_b h^2$.
ω_c / omch^2	Physical cold dark matter density $\Omega_c h^2$.

Variable	Meaning
N_eff / nnu	Effective relativistic species count inferred independently in the CMB nuisance chain.
Y_p / YHe	Primordial helium-4 mass fraction; sampled independently in the full CMB likelihood.
D/H	Deuterium abundance anchor; required to remain within the adopted observational gate.
Li-7/H	Lithium-7 abundance. Most primordial mass-7 is produced as Be-7 and later decays to Li-7.
m_phi	Mediator mass in MeV for the selective electromagnetic-transfer candidate.
E_gamma	Characteristic photon energy, approximated as m_phi/2 for two-photon decay.
tau_phi	Mediator lifetime in seconds.
n_phi/n_gamma	Mediator abundance relative to photons at T = 10 MeV.
g_phi_gamma	Effective mediator-photon coupling in GeV^-1.
mu	CMB spectral-distortion chemical-potential style readout.
R-1	Convergence diagnostic used by Cobaya and the split-chain audit.
chi2	Goodness-of-fit statistic; lower values indicate a better fit for the same data and model complexity.
A(Li)	Logarithmic stellar lithium abundance, A(Li) = log10(Li/H) + 12.
Y_p	Primordial helium-4 mass fraction used by named helium abundance anchors.
tau_int	Integrated autocorrelation time used in the strict posterior qualification audit.
ESS	Effective sample size; the approximate number of independent posterior draws after autocorrelation.
q_i	Dimensionless pull applied to reaction i in the registered D-chain response parameterisation.
f_i(T9)	Temperature-dependent multiplicative one-sigma factor uncertainty for reaction i.
delta_i(T9)	Physical fractional rate shift exp[q_i ln f_i(T9)] - 1.
D_M^2	Squared Mahalanobis distance delta^T C+ delta from the external rate posterior mean.
T9	Temperature in units of 10^9 kelvin.

Appendix C. Program and Data Manifest

Table C1. Principal programs

Program or path	Role in the investigation
analyze_bbn_lithium_linx_fr_network_2026-07-16.py	July LINX/FR network controls; reproduced the lithium problem and did not identify a D+He+Li pass.
analyze_su2_bbn_lithium_gate_2026-07-17.py	July SU2-style expansion/thermal gate for lithium; part of the negative control phase.
validate_bbn_lithium_linx_matrix_2026-07-17.py	Strict July validation matrix for the BBN lithium controls.
audit_bbn_lithium_linx_fr_scan_2026-07-17.py	Audit layer for the July LINX/FR scan.
run_cobaya_spd_cache_builder_2026_08_21.py	Built the validated SPD cache for the Planck PR4/NPIPE CamSpec likelihood.
postprocess_planck_pr4_full_nuisance.py	Post-processed the native full CMB nuisance chain and compared the registered lithium cell with the CMB posterior.
audit_planck_pr4_full_nuisance_qualification.py	Strict qualification audit: split chains, autocorrelation-aware ESS, registered-cell density, nuisance stability and half-chain stability.
start_planck_pr4_full_nuisance_background.ps1	Background launcher for the native Planck PR4 full nuisance MCMC.
start_planck_pr4_full_nuisance_ess_topup_background.ps1	Background launcher for the ESS top-up chain.
build_bbn_li_reports_20260911.py	This reproducible DOCX report generator.
run_bbn_lithium_nlo_thermal_2026_09_11.py	September matched NLO thermal gate for the registered mediator cell; checks pole-cut consistency, thermal self-energy proxies and NLO-size stability.
run_bbn_lithium_anchor_sensitivity_2026_09_11.py	September generic sensitivity grid over D/H, helium, stellar lithium and nuclear-rate proxy choices.
build_bbn_li_named_anchor_table.py	September named literature-anchor table and stress grid linking the registered candidate to specific observational anchors.

Program or path	Role in the investigation
github_export/code/bbn_lithium/run_bbn_li_primat_rate_mc.py	PRIMAT small-network rate Monte Carlo importer and covariance bridge runner, including the N=5000 validated-response run.
github_export/code/bbn_lithium/diagnose_bbn_li_linx_covariance_eigenmode_validation.py	Direct LINX reruns along PRIMAT-derived covariance eigenmodes to validate local response linearity.
github_export/code/bbn_lithium/audit_bbn_li_covariance_stability_failure.py	N=1000 versus N=5000 covariance stability and abundance-failure decomposition.
github_export/code/bbn_lithium/diagnose_bbn_li_atlas_covariance_proxy.py	BBN sensitivity-atlas and NACRE-II covarianceproxy stress diagnostic.
github_export/code/bbn_lithium/diagnose_bbn_li_direct_cross_network.py	Direct abundance comparison between exact LINX points and matched PRIMAT reruns.
github_export/code/bbn_lithium/postprocess_bbn_li_direct_ddchain_worst_points.py	Worst-direct-point post-process for the fresh Dchain LINX/PRIMAT validation run.
diagnose_bbn_li_galah_hierarchical_depletion.py	Censored GALAH DR3 stellar-depletion/plateau model with a frozen GALAH DR4 validation.
rescore_bbn_li_archive_with_galah_posterior.py	Non-destructive archive-wide reclassification under the qualified GALAH lithium posterior.
diagnose_bbn_li_adaptive_shotgun.py	Adaptive 200,000-row mediator plus D-chain hypothesis search; not a confirmation sampler.
diagnose_bbn_li_combined_mediator_dchain.py	Direct LINX operator-split propagation of selected mediator plus D-chain candidates.
diagnose_bbn_li_candidate_external_covariance.py	External D-chain covariance audit and sourceavailability register.
update_bbn_li_problem_paper_2026_09_25_latest_results.py	Reproducible builder for this 25 September manuscript revision.
run_bbn_li_primat_nlo_fullfluence.py	Runs the 10,000-draw full-fluence-matched NLO-proxy robustness branch.
diagnose_bbn_li_fullfluence_linx_validation.py	Performs stratified direct LINX source validation with segmented convergence control.
diagnose_bbn_li_action_lifetime_shutoff.py	Derives the conditional scalar lifetime/dilution history and tests source shut-off responses.
run_bbn_li_acropolis_mediator.py	Runs the exact ACROPOLIS 1.3.1 non-universal decay-cascade parameter scan.
diagnose_bbn_li_acropolis_cosmology_closure.py	Replays exact pass rows against modern PRIMAT, production, eta/N_eff and FIRAS screens.
diagnose_bbn_li_parent_decay_acropolis_bridge.py	Bridges the action-conditioned parent source into native ACROPOLIS evolution.
diagnose_bbn_li_independent_line_cascade.py	Implements the independent reduced line-cascade candidate check.
run_bbn_li_independent_cascade_basin_overnight.py	Extends reduced-cascade comparison across all 58 exact ACROPOLIS pass rows.
diagnose_bbn_li_explicit_pair_cascade.py	Adds angle-integrated Breit-Wheeler pair production and secondary-energy closure.
diagnose_bbn_li_active_band_cascade.py	Solves the Be7-active photon-band redistribution recurrence for boundary rows.
diagnose_bbn_li_basin052_boundary_robustness.py	Audits GALAH ESS and mass-7 cross-section sensitivity for the remaining control.

Table C2. Repository and output cross-reference

Object	Path or URL
Local output root	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821
Combined qualification audit	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\qualification_audit_combined_ess_topup
Post-process folder	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\postprocess
GitHub project	https://github.com/cliveboyd/Cosmology
GitHub BBN code	https://github.com/cliveboyd/Cosmology/tree/main/github_export/code/bbn_lithium
GitHub papers	https://github.com/cliveboyd/Cosmology/tree/main/github_export/docs/papers
September NLO thermal gate	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_nlo_thermal_gate_20260911
September generic anchor sensitivity	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_anchor_sensitivity_20260911
September named anchor sensitivity	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_named_anchor_sensitivity_20260911
N=5000 PRIMAT covariance bridge	plamb_runs/diagnostics/bbn_li_primat_rate_mc/bbn_li_primat_rate_mc_2026-0912_primat_small_N5000_seed913_validated_response_report.md
N=5000 direct LINX eigenmode validation	plamb_runs/diagnostics/bbn_li_linx_eigenmode_validation/bbn_li_linx_eigenmode_validation_2026-0912_small12_N5000_top8_pm1_pm2_report.md
Covariance stability and D/H failure audit	plamb_runs/diagnostics/bbn_li_covariance_stability_failure_audit/bbn_li_covariance_stability_failure_audit_2026-09-12_report.md
Atlas covariance-proxy stress test	plamb_runs/diagnostics/bbn_li_atlas_covariance_proxy/bbn_li_atlas_covariance_proxy_2026-0912_atlas_response_proxy_N5000000_report.md

Object	Path or URL
Fresh D-chain direct crossnetwork validation	plamb_runs/diagnostics/bbn_li_direct_cross_network/bbn_li_direct_cross_network_2026-0912_fresh_ddhe3n_ddtp_overnight_report.md
Fresh D-chain worst-point post-process	plamb_runs/diagnostics/bbn_li_direct_cross_network/bbn_li_direct_cross_network_2026-0912_fresh_ddhe3n_ddtp_overnight_worst_points_summary.md
GALAH hierarchical report	plamb_runs/diagnostics/bbn_li_galah_hierarchical_depletion/galah_hierarchical_depletion_validation_report.md
GALAH archive re-score	plamb_runs/diagnostics/bbn_li_galah_archive_rescore/bbn_li_galah_archive_rescore_report.md
GALAH 200k adaptive search	plamb_runs/diagnostics/bbn_li_adaptive_shotgun/bbn_li_adaptive_shotgun_2026-0923_galah_posterior_extensive_200k_s1_report.md
Direct 60-row LINX batch	plamb_runs/diagnostics/bbn_li_combined_mediator_dchain/bbn_li_combined_mediator_dchain_2026-0923_galah_posterior_extensive60_safebatch_s2_report.md
External covariance audit	plamb_runs/diagnostics/bbn_li_candidate_external_covariance/bbn_li_candidate_external_covariance_2026-0924_galah60_external_covariance_report.md

Exact ACROPOLIS mediator scan	plamb_runs/diagnostics/bbn_li_acropolis_mediator/
Action and production closure	plamb_runs/diagnostics/bbn_li_action_lifetime_shutoff/; bbn_li_parent_decay_acropolis_bridge/
Independent cascade sequence	plamb_runs/diagnostics/bbn_li_independent_line_cascade/; bbn_li_independent_cascade_basin/; bbn_li_explicit_pair_cascade/
Active-band and boundary audit	plamb_runs/diagnostics/bbn_li_active_band_cascade/; bbn_li_basin052_boundary_robustness/
GitHub October result bundle	github_export/results/2026-10-02/bbn_lithium/

Internal reproducibility references for the September 12 nuclear-rate update are listed in Appendix C and in the BBN research-hub register entries BBN-013 through BBN-022.

Appendix D1. External Reference Links Used in the Analysis

The following links are included so the publication draft connects each major claim boundary to an external source or numerical resource.

Reference	Link	Analytical use
Burns 2026 Physics Reports / BBN sensitivity atlas	Open source	The BBN-Li paper should mirror this logic: each external reference defines a gate or comparison, not merely a bibliography entry.
arXiv:2603.22414	Open source	Provides a direct model for linking reaction sensitivities and uncertainty budgets to the lithium problem.
BBN sensitivity atlas GitHub	Open source	Used as the main external response benchmark for the September 12 atlas import and NACREII stress branch.
Fields 2011 lithium review	Open source	Frames why a successful mechanism must reduce mass-7 without breaking the otherwise successful BBN light-element pattern.
Cybert Fields Olive Yeh 2016	Open source	Defines why broad expansion or baryon-density retuning is a high-risk route and why D/H and helium gates are non-negotiable.
Pitrou Coc Uzan Vangioni 2018 PRIMAT	Open source	Supports keeping PRIMAT cross-network comparison separate from a purely local LINX response result.
LINX code paper	Open source	Supports the local response-derivative/eigenmode-validation strategy, while still requiring direct mediatoraware propagation.
LINX arXiv preprint	Open source	Supports the technical decision to use LINX for fast response scans and direct finite-pull reruns.
ACROPOLIS electromagnetic-cascade framework	Open source	Defines why a late electromagnetic-transfer candidate needs cascade and threshold gates, not just a lithium-yield fit.
Planck 2018 cosmological parameters	Open source	Motivates replacing compressed CMB priors with the native PR4/NPIPE nuisance posterior used in September.
COBE FIRAS CMB spectrum	Open source	Defines the need for spectral-safety gates for any post-BBN electromagnetic energy release.
Khatri and Sunyaev 2012 CMB spectrum formation	Open source	Frames why mu-distortion and thermalisation checks are part of the candidate safety case.
CosmoVerse White Paper 2025	DOI	Added as an external gate reference: lithium improvement must survive D/H, Yp, He3/D, CMB, spectral, nuclear-rate and cross-dataset checks before promotion.
Wang et al. 2024 GALAH 3D-NLTE lithium catalogue	Open source	Large stellar-lithium catalogue used to build the empirical censored plateau/depletion model.
Launders, Giovanetti and Liu 2026 deuterium GP analysis	Open source	Highest-priority source for synchronised, correlated D-chain functional uncertainty.
Pisanti et al. 2021, Primordial Deuterium after LUNA	Open source	Published coefficient/error-budget basis used for the executable partial external covariance gate.
Mossa et al. 2020 LUNA deuterium-burning	Open source	Experimental d(p,γ)3He anchor

Reference	Link	Analytical use
measurement		underlying the improved deuterium-rate programme.
Gomez Inesta, Iliadis and Coc 2017 Bayesian d+d rates	Open source	Bayesian treatment of the paired d(d,n) ³ He and d(d,p)t reaction rates.
Moscoso et al. 2021 Bayesian d(p,γ) ³ He rate	Open source	Independent hierarchical Bayesian rate analysis for the third D-sensitive reaction.

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